

THOMAS PULICKAL JOSEPH

SENIOR SOFTWARE ENGINEER

WEST CHESTER, PA

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Professional Summary

Senior Software Engineer with experience building high-throughput automation platforms, real-time control systems, and distributed software solutions for consumer electronics manufacturing. Specialized in C#, ASP.NET Core, Beckhoff TwinCAT, robotics integration, and scalable backend systems. Proven track record designing production-grade automation software, concurrent communication frameworks, and telemetry platforms that improve throughput, reliability, and operational efficiency.

Experience

COMMUNICATIONS TEST DESIGN, INC

SENIOR SOFTWARE ENGINEER

JAN 2026 – PRESENT
West Chester, PA

- Developed a real-time C# communication framework using the Task Parallel Library (TPL) to bridge enterprise software and industrial hardware, sustaining over 1,000 communication transactions per second through TwinCAT ADS/AMS integration.
- Designed and developed internal web applications hosted on internal company infrastructure using HTML, PHP, and SQL to support operations.
- Built dynamic data-driven web pages with PHP backend and SQL database integration for real-time production data access.
- Created internal tooling and dashboards for test result visualization and reporting via web interfaces.
- Architected and deployed high-throughput automated test cells for consumer electronics, specifically integrating Beckhoff xPlanar motion platforms with custom fixtures to eliminate manual handling.
- Engineered complex motion sequences using TwinCAT 3 (Structured Text), optimizing mover routing and precision positioning for synchronized electrical signaling and power-on testing.
- Architected and deployed cross-platform telemetry and control systems using Avalonia UI and ASP.NET Core, enabling real-time diagnostics, automated grading workflows, and production monitoring.
- Analyzed robotic motion profiles and sensor data to refine cycle times, resulting in a 30% increase in process throughput and a 40% reduction in system downtime.
- Designed and implemented RESTful APIs to synchronize robotic test results with enterprise databases.
- Collaborated cross-functionally with electrical and mechanical engineering teams to translate hardware specifications into scalable software solutions, utilizing Git workflows and Octopus Deploy for controlled production releases.
- Engineered machine-vision inspection systems for automated display grading, utilizing advanced image processing to detect micro-defects and pixel irregularities on mobile device screens, significantly increasing grading consistency.
- Optimized PLC communication pipelines and multithreaded concurrency handling across robotic cells to ensure fail-safe performance in high-volume manufacturing environments.
- Authored comprehensive technical documentation, including system diagrams and test procedures, to facilitate seamless maintenance and continuous improvement cycles.

SOFTWARE ENGINEER

AUG 2023 – DEC 2025
West Chester, PA

- Developed modular automation platforms for device handling and semi-robotic operations using C#, ASP.NET Core, and Knockout.js to support CTDI test and repair workflows.
- Developed and maintained web-based phone testing software used across CTDI operations, integrating automation platforms, test equipment, and backend services to support high-volume device processing.
- Designed and maintained web-based control dashboards for real-time visibility into automation status, queues, and device-level test results.
- Implemented REST APIs to integrate automation modules, test sequencers, and internal/third-party systems, enabling reliable data exchange and orchestration of test flows.
- Supported server-side automation using Apache HTTP Server with PHP and SQL for processing test data, generating reports, and monitoring system health.
- Improved stability, efficiency, and maintainability by refactoring legacy services, adding automated validation, and enhancing observability.
- Collaborated in Agile teams using Git and Jira for source control, code review, issue tracking, and controlled deployment of releases into CTDI production environments.

MPUNDIT SOLUTIONS PVT LTD

SOFTWARE ENGINEER

SEP 2020 – JULY 2021
Guntur, India

- Developed microservices and device-control APIs for industrial automation systems using C# .NET and PostgreSQL.
- Integrated backend logic for machine-assisted monitoring, improving operational precision and speed.
- Designed optimized SQL schemas and procedures for high-performance analytics.
- Contributed to full-stack projects in Agile Scrum environments for timely delivery of automation software.

- Built a hospital management system featuring automated patient record handling in PHP and MongoDB.
- Streamlined data exchange workflows and improved user-facing interface reliability.

Technical Skills

- **Programming Languages:** C#, C++, Python, Structured Text (PLC), Java, C, PHP, SQL.
- **Robotics & Industrial Automation:** Beckhoff TwinCAT 3, EtherCAT, xPlanar Motion Control, Real-Time Distributed Systems.
- **Electrical & Electronics:** Digital/analog I/O, sensor and actuator integration, custom test fixture integration and Electrical Test Sequencing.
- **Computer Vision:** Vision-driven inspection for functional grading.
- **Backend & Web Technologies:** ASP.NET Core, Avalonia UI (Cross-platform XAML), JavaScript, Knockout.js, ReactJS, PHP, Apache HTTP Server, RESTful API Development.
- **DevOps & Tools:** Git, GitHub, Octopus Deploy, Postman, Jira, Visual Studio, TwinCAT Scope View.
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB.
- **Methodologies:** Agile, Scrum, CI/CD, SDLC, TDD (Test Driven Development)

Education

UNIVERSITY OF HOUSTON-CLEAR LAKE
Master of Science in Computer Science (GPA: 3.70 / 4.00)

May 2023
Houston, TX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY
Bachelor of Technology in Computer Science & Engineering (GPA: 7.75 / 10.00)

Sep 2020
Hyderabad, India

Certifications

Beckhoff TwinCAT 3 Robotics Training (TR3025)

Basic I/O, Communication, System Manager and PLC Programming (Beckhoff Automation, Feb 2025)

IBM Data Science Professional Specialization

Python, data analytics, visualization, and machine learning fundamentals (IBM / Coursera, Jul 2022)

Joy of Computing using Python

Python programming, algorithms, and computational thinking (SWAYAM MHRD / NPTEL, Dec 2019)